2D Rasterisation: Lines, Triangles and a House

COMP3033J Computer Graphics — Assignment 1
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A. Theory

An explicit line is $y = mx + b$, with $m = (y_1 - y_0) / (x_1 - x_0)$ and $b = y_0 - mx_0$. Vertical lines have undefined slope; steep lines require sampling along y.

The implicit form $F(x, y) = Ax + By + C = 0$ does not choose a dependent variable. Here $A = y_0 - y_1$, $B = x_1 - x_0$ and $C = x_0y_1 - x_1y_0$. The perpendicular distance is given below; the sign of F indicates the side of the directed line. A pixel is drawn when d is below a small tolerance. The same side test, repeated for three edges, fills a triangle.

$$d = \frac{|Ax + By + C|}{\sqrt{A^2 + B^2}}$$

The edge function $E_{AB}(P) = (B_x - A_x)(P_y - A_y) - (B_y - A_y)(P_x - A_x)$ is twice the signed area of ABP. P is inside when $E_{BC}(P)$, $E_{CA}(P)$ and $E_{AB}(P)$ share the sign of the area of ABC. Following lecture 05, signed distance ratios give $\alpha = d(P,BC)/d(A,BC)$ and $\gamma = d(P,AB)/d(C,AB)$; $\beta = 1 - \alpha - \gamma$. Distances are to the infinite directed edge lines and retain their signs. These ratios equal the signed area ratios because the triangles share a base. They sum to 1, and P is inside when all three are non-negative. Colour is the linear blend $\alpha C_A + \beta C_B + \gamma C_C$.

B. Algorithms

Explicit lines. A vertical segment is drawn by stepping in y at fixed x. Otherwise the code compares $|\Delta y|$ with $|\Delta x|$. A shallow line is ordered so x increases, and each integer x receives $y = mx + b$. A steep line is ordered in y, and $x = ny + d$, where $n = \Delta x/\Delta y$ and $d = x_0 - ny_0$, follows the lecture's alternative explicit form. setPixel rounds to the nearest pixel. The branch only chooses which axis to sample.

Implicit lines follow lecture 05, page 7: scan a clipped bounding box and test $|distance| < 0.5$. Projection $t = ((P-A)\cdot(B-A))/|B-A|^2$ limits the strip to [0,1]. Rounded endpoints preserve short segments. At distance 0.5, a fixed coefficient sign selects one side independently of endpoint order; a 10^{-10} tolerance recognises ties.

Half-plane fill follows page 23. A negative signed area swaps B and C, then three edge-function sign tests implement the lecture's leftOf checks. This is algebraic positive winding; it appears clockwise on the y-down screen. Both triangle modes sample integer (x,y), matching the lecture pseudocode. Zero-area triangles are skipped. Shared edges are included.

Barycentric fill follows page 29: calculate the distance ratios, reject negative weights and blend RGB. A 10^{-10} weight tolerance absorbs round-off at edges. The standalone weight function returns null for a degenerate triangle. Rectangles use two triangles. The sun is twelve shaded triangles around a shared centre, forming a faceted twelve-sided polygon. All scenery uses lines and triangles. Babylon.js only displays the buffer.

C. Comparison

The explicit equation is the easiest to read, because it is $y = mx + b$. It is the least convenient to rasterise. Vertical lines need a separate loop, steep lines break up unless the major axis changes, while choosing the major axis keeps the sampled slope magnitude at most one.

The implicit test uses one expression for every direction. Implementation is a double loop, which is short but visits many empty pixels. F scales with segment length, so the comparison must use distance rather than raw F. The half-pixel test and rounding can choose different pixels. It provides binary coverage rather than anti-aliasing.

Barycentric weights provide both triangle coverage and colour interpolation. Nearly degenerate triangles can amplify floating-point error; exactly degenerate ones are skipped.

D. Results

Experiment 1 compares orientation. Figures 1 and 2 contain the same six segments: shallow (black), horizontal (red), diagonal (blue), steep (green), vertical (purple) and a second steep line (dark grey). Explicit sampling along y keeps the steep and vertical lines continuous; without that branch they would be dotted. The implicit image follows the same geometry with a half-pixel distance test. Both methods show stair steps on diagonals because coverage is binary. Horizontal and vertical examples remain one pixel wide; projection bounds the interior samples. A regression test at $y = 5.5$ confirms one continuous row: the lecture's strict test alone would reject both neighbouring rows.

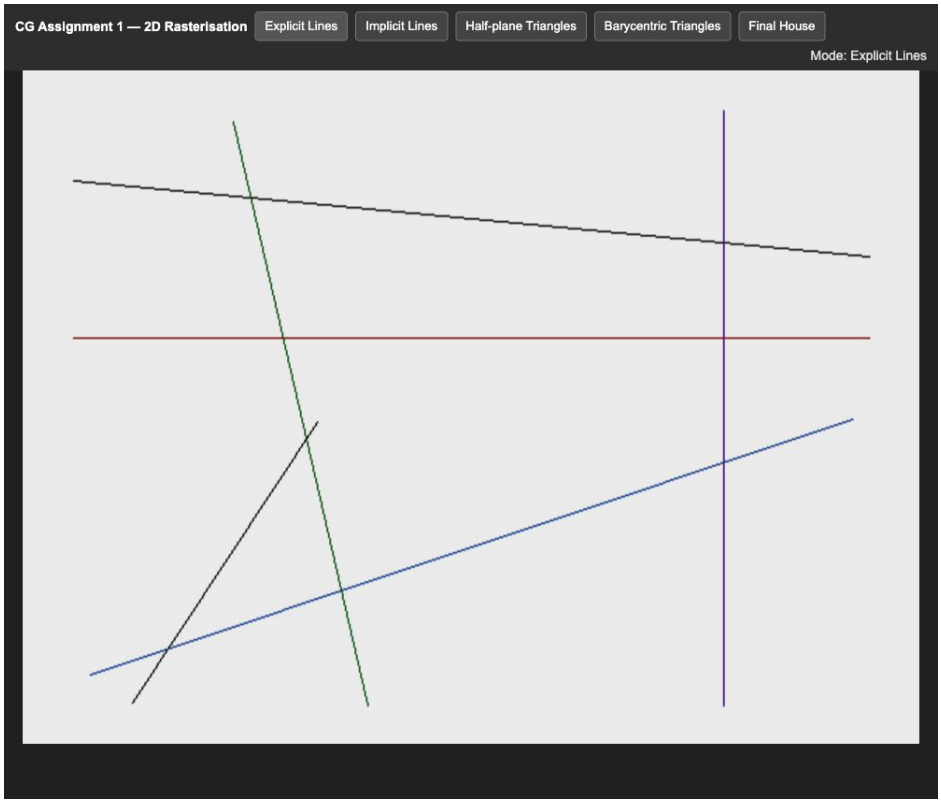


Figure 1. Explicit lines across shallow, steep, horizontal, vertical and diagonal cases.

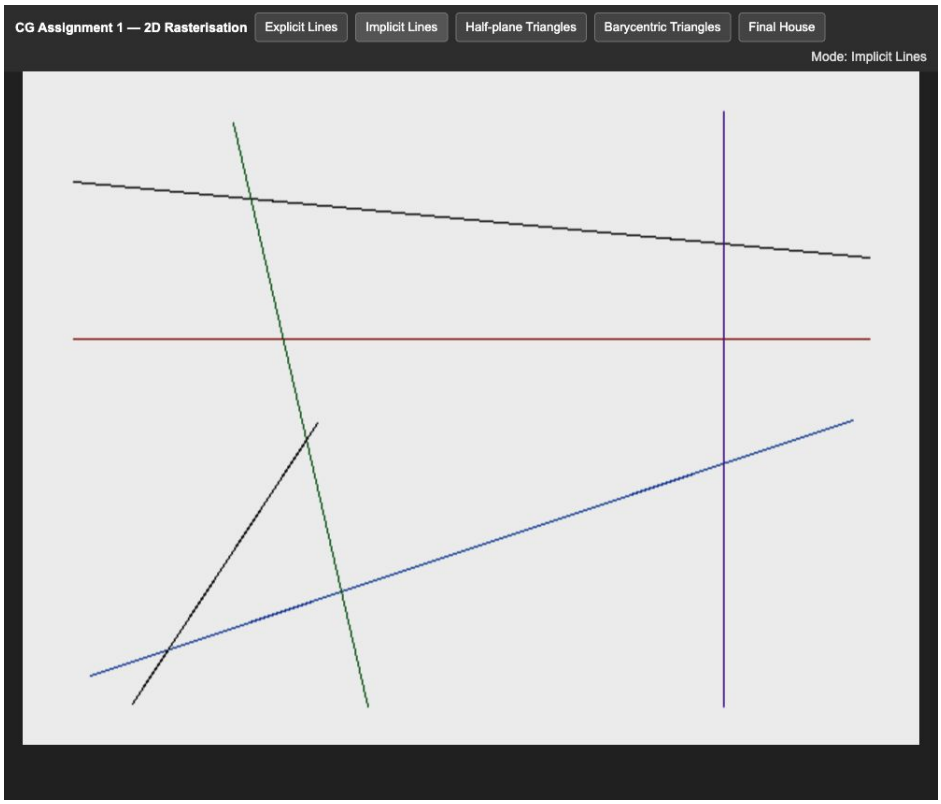


Figure 2. The same segments under the implicit distance test.

Experiment 2 is triangle coverage (Figure 3). The blue triangle has a horizontal edge and a vertical edge. The orange triangle is acute. The green triangle is obtuse at its lower vertex. None of these shapes needs a special fill rule. An integer sample is painted only when it is inside every edge half-plane. Axis-aligned edges use the same test. Implicit lines then outline each triangle.

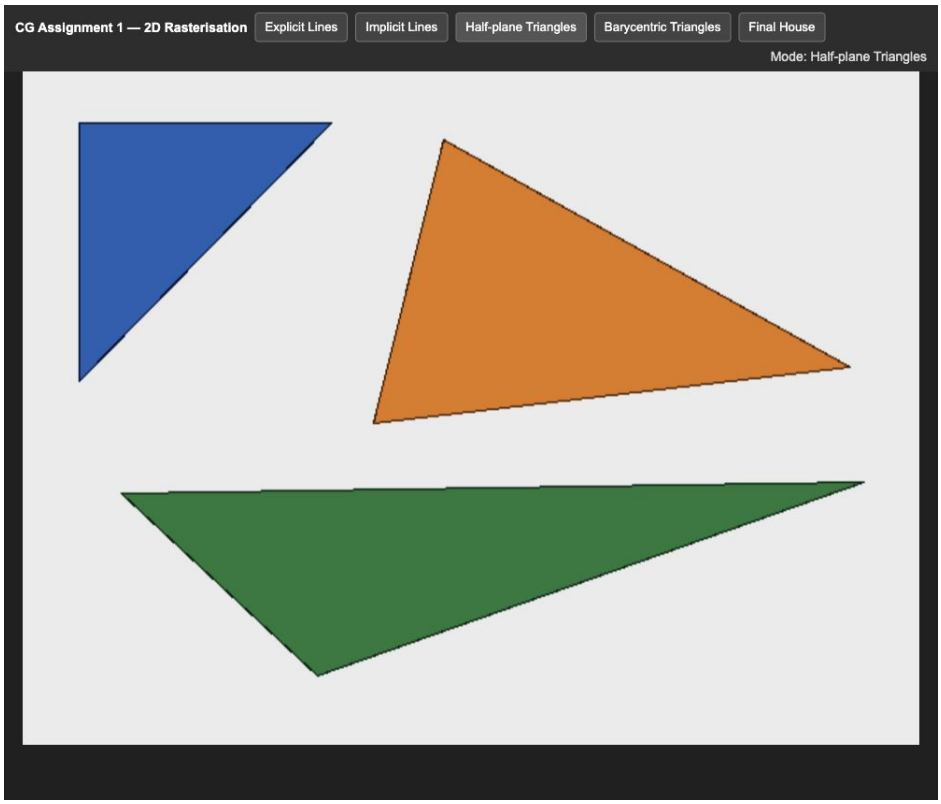


Figure 3. Half-plane fill of a right triangle, an acute triangle and an obtuse triangle.

Experiment 3 uses strongly different vertex colours (Figure 4): red, green and blue on the left, and yellow, magenta and cyan on the right. At a vertex one weight is 1 and the others are 0, so the colour matches that vertex. Around the centroid each weight is near one third, and the left triangle turns a dark mix of the three primaries. On an edge the opposite weight is 0, so only the two endpoints are blended. The change is smooth because each weight is linear in x and y.

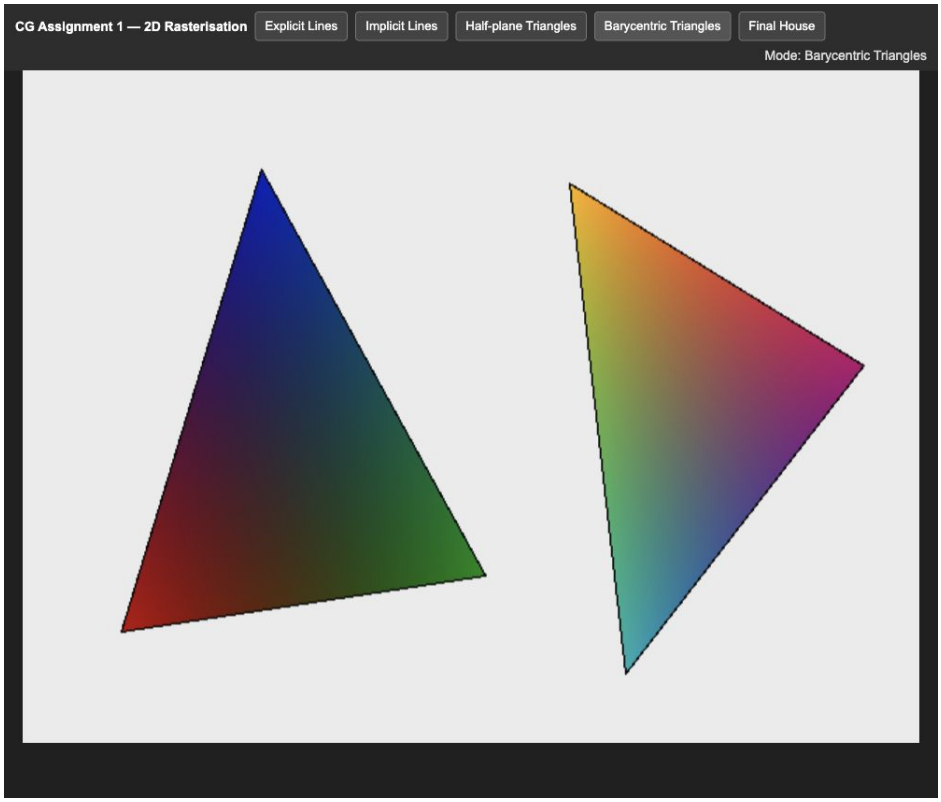


Figure 4. Colour interpolated from three different vertex colours.

Figure 5 is the house required by the final task: a rectangular body, a triangular roof, a door, windows, a chimney and straight decorative lines. Walls and the roof are half-plane triangles. Sky, glass and the sun use barycentric colour. The Babylon.js plane in main.js only holds the texture; it is not the house geometry.

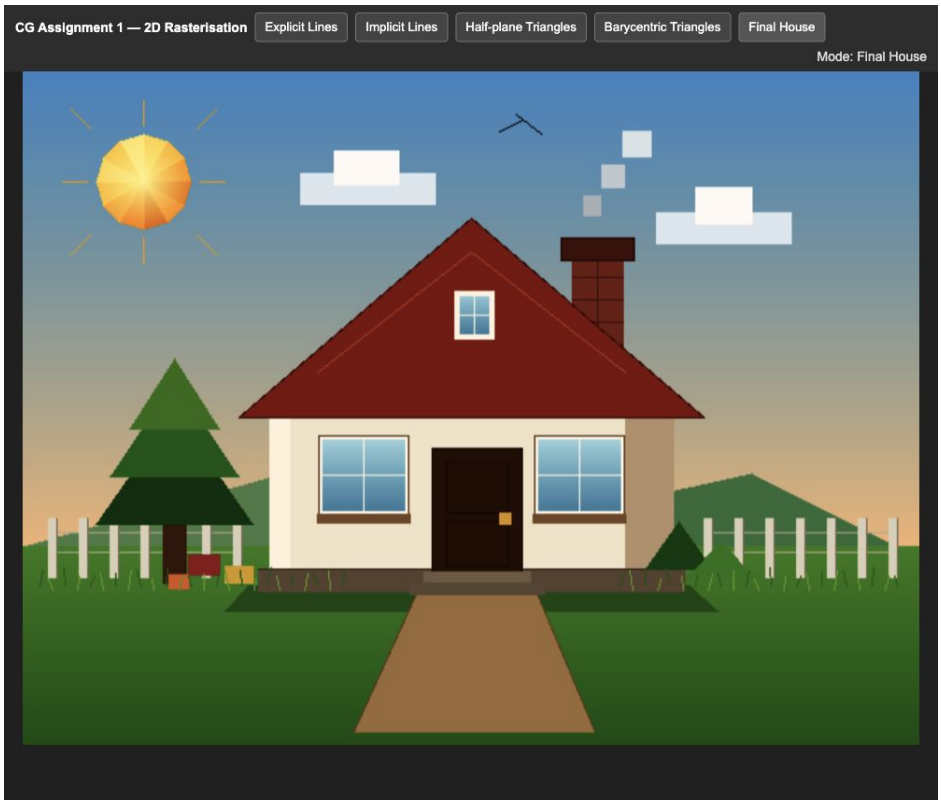


Figure 5. The house and faceted sun produced by the line and triangle rasterisers.

E. Reflection

Rasterisation asks whether a sample belongs to a primitive. Explicit equations generate positions but need orientation-dependent branches. Half-plane tests answer coverage directly using three edge signs. Barycentric coordinates divide signed edge distances by the corresponding vertex distances, providing both coverage and colour weights. They therefore extend naturally to shaded triangles. Bounding-box scanning is simple but inefficient for thin primitives; incremental methods would reduce repeated calculations.